

A Hierarchical Gene-Isoform Model for Sparse Single-Cell Data

Purpose

The existing genome-wide factorization models normalized equivalence-class (EC) counts directly. It recovers cell-type information, but a matched evaluation shows a substantial gap from a standard gene-expression analysis. Increasing the factor rank does not close this gap. Adding an auxiliary gene-reconstruction loss helps, indicating that the EC objective spends too much capacity on transcript ambiguity and too little on marker-gene abundance.

This note proposes a hierarchical count model that separates gene abundance from conditional isoform allocation. The model is designed for cells that do not contain enough molecules for independent transcript-level EM. Statistical strength instead comes from pooled transcript baselines, low-dimensional cell states, and transcript loadings shared across all cells.

Notation

Let $m = 1, \dots, M$ index cells, $g = 1, \dots, G$ genes, and $t = 1, \dots, T$ transcripts. Let $g(t)$ be the gene containing transcript t , and let $\mathcal{T}_g = \{t : g(t) = g\}$. Primer or library type is indexed by p . For primer p , let

$$A_p \in \mathbb{R}_+^{K_p \times T}$$

be the fixed EC-by-transcript observation matrix. It includes alignment compatibility, Salmon conditional weights and positional-bias correction, and the primer-specific transcript sampling exposure. In particular, the primary paired model uses no additional transcript-length exposure for oligo(dT) and relative effective-length exposure for random-hexamer priming.

Let n_{kmp} be the observed UMI count for EC k , cell m , and primer p , and let $N_{mp} = \sum_k n_{kmp}$.

Hierarchical Abundance

Transcript abundance is decomposed as

$$\theta_{tm} = a_{g(t)m} \pi_{tm}, \quad \sum_{t \in \mathcal{T}_g} \pi_{tm} = 1.$$

Here $a_{gm} > 0$ is the abundance of gene g in cell m , while π_{tm} is the conditional fraction of gene $g(t)$'s molecules assigned to transcript t .

This within-gene normalization is not a global simplex constraint on Θ . It is needed to identify gene abundance separately from isoform scale. The gene abundances may remain unnormalized because the observation likelihood below normalizes predicted EC mass within each cell and primer.

Gene abundance factors

Let $\mathbf{z}_m \in \mathbb{R}^{r_g}$ be the gene-expression state of cell m . Model gene abundance as

$$\log a_{gm} = \alpha_g + \mathbf{u}'_g \mathbf{z}_m,$$

where α_g is a pooled gene intercept and $\mathbf{u}_g$ is a gene loading. The cell state $\mathbf{z}_m$ is the natural representation for cell annotation, visualization, and comparison with standard gene-expression analyses.

An exponential link is convenient for a count model. A softplus link can be used if bounded gradients are more stable:

$$a_{gm} = \text{softplus}(\alpha_g + \mathbf{u}'_g \mathbf{z}_m).$$

Conditional isoform programs

For gene g , let

$$P_g \in \mathbb{R}_+^{T_g \times C_g}, \quad \mathbf{1}' P_g = \mathbf{1}',$$

where $T_g = |\mathcal{T}_g|$. Each column of P_g is a conditional isoform usage program, and C_g is small, typically one to three. Let $\mathbf{s}_{gm} \in \mathbb{R}_+^{C_g}$ be the cell-specific program weights:

$$\lambda_{gmc} = \kappa_{gc} + \mathbf{v}'_{gc} \mathbf{z}_m, \quad s_{gmc} = \frac{\exp(\lambda_{gmc})}{\sum_{d=1}^{C_g} \exp(\lambda_{gmd})}.$$

Conditional transcript usage is

$$\boldsymbol{\pi}_{gm} = P_g \mathbf{s}_{gm}.$$

Thus the same robust cell state $\mathbf{z}_m$ controls gene abundance through $\mathbf{u}_g$ and isoform-program weights through $\mathbf{v}_{gc}$. A cell does not receive an unrestricted transcript vector.

If held-out molecules support additional isoform structure, introduce a smaller isoform-specific state $\mathbf{h}_m \in \mathbb{R}^{r_i}$:

$$\lambda_{gmc} = \kappa_{gc} + \mathbf{v}'_{gc} \mathbf{z}_m + \mathbf{r}'_{gc} \mathbf{h}_m.$$

The $\mathbf{r}'_{gc} \mathbf{h}_m$ term captures program changes that are not predictable from gene expression. The nested model $r_i = 0$ is the default, and $r_i > 0$ is accepted only by label-blind molecule-count cross-validation.

Restriction relative to a flat model

The decomposition $\theta_{tm} = a_{g(t)m} \pi_{tm}$ alone is only a reparameterization. Given any positive $\boldsymbol{\theta}_m$, its gene totals and within-gene proportions recover a_{gm} and $\boldsymbol{\pi}_{gm}$. A multinomial likelihood does not make an unrestricted hierarchical model more informative than a flat model.

The proposed model gains statistical strength from its restrictions:

- gene abundance is generated by the low-dimensional state $\mathbf{z}_m$;
- each P_g is shared by every cell;
- $C_g \ll T_g$ when a gene has several transcripts;
- program weights are generated by $\mathbf{z}_m$ and, optionally, a small $\mathbf{h}_m$, rather than estimated independently in every cell.

If $C_g = T_g$, P_g is the identity, and $\mathbf{s}_{gm}$ is unrestricted, the model reduces to the flat multinomial parameterization. That limiting case is not the proposed sparse-cell model.

Connection to the Paired-Primer Likelihood

The conceptual transcript abundance remains

$$\theta_{tm} = a_{g(t)m}[P_{g(t)}\mathbf{s}_{g(t)m}]_t,$$

but it need not be instantiated. Project each gene’s shared programs through the primer-specific observation model:

$$D_{pkgc} = \sum_{t \in \mathcal{T}_g} A_{pkt} P_{gtc}.$$

The unnormalized expected EC mass is then exactly

$$q_{kmp} = \sum_g \sum_{c=1}^{C_g} D_{pkgc} a_{gm} s_{gmc}.$$

This equation is the direct connection between the gene factors, isoform factors, and observed EC counts. It handles within-gene and cross-gene ambiguous ECs because all genes represented in row k contribute to the sum.

Conditioning on the observed UMI total gives

$$\mathbf{n}_{mp} \mid N_{mp} \sim \text{Multinomial} \left(N_{mp}, \frac{\mathbf{q}_{mp}}{\mathbf{1}'\mathbf{q}_{mp}} \right).$$

Ignoring constants, the negative log likelihood is

$$\mathcal{L} = - \sum_{m,p} \sum_{k: n_{kmp} > 0} n_{kmp} \log q_{kmp} + \sum_{m,p} N_{mp} \log(\mathbf{1}'\mathbf{q}_{mp}).$$

For the normalizer, precompute

$$\delta_{pgc} = \sum_k D_{pkgc} = \sum_{t \in \mathcal{T}_g} d_{pt} P_{gtc}, \quad d_{pt} = \sum_k A_{pkt}.$$

It follows that

$$\mathbf{1}'\mathbf{q}_{mp} = \sum_g \sum_c \delta_{pgc} a_{gm} s_{gmc}.$$

Neither likelihood term requires a transcript-by-cell abundance block.

The two primer halves share $\mathbf{z}_m$, $\mathbf{h}_m$, a_{gm} , and $\mathbf{s}_{gm}$, but use separate D_p obtained from their separate A_p . Conditioning on N_{mp} removes a primer-by-cell exposure parameter and preserves the current convention of normalizing each half independently.

An independent Poisson model,

$$n_{kmp} \sim \text{Poisson}(s_{mp} q_{kmp}),$$

is an alternative when total UMI information is intended to contribute to fitting. It requires estimating or profiling the exposure s_{mp} . The conditional multinomial model is the simpler first implementation and directly matches the sampling unit used for molecule-count folds.

Why Cell-Level EM Is Not Required

Independent EM would estimate a high-dimensional transcript vector in every cell. That is underdetermined for cells with hundreds or a few thousand UMIs. The hierarchical model instead estimates only low-dimensional cell coordinates:

$$\mathbf{z}_m \in \mathbb{R}^{r_g}, \quad \mathbf{h}_m \in \mathbb{R}^{r_i}.$$

All gene and program parameters are shared across cells. Cells without molecules that distinguish isoforms remain close to the pooled program mixture given by P_g and κ_{gc} . Informative cells update the shared relationship between cell state and program usage.

EM may still be useful on pooled data to initialize the columns of P_g . A global or pseudobulk EM estimate has enough molecules to establish baseline isoform usage, but it is not treated as a cell-level estimator. Joint minibatch gradient optimization of the count likelihood then estimates cell states, program weights, and shared programs.

Initialization and Fitting

A staged fit reduces non-convexity and provides useful diagnostics:

1. Aggregate ECs that map entirely within one gene and fit $\alpha_g, \mathbf{u}_g, \mathbf{z}_m$ with a gene-level count factor model. Cross-gene ambiguous ECs are excluded only from this initializer.
2. Initialize one column of P_g from pooled EM. Additional columns can be initialized from large pseudobulks or small perturbations of the pooled profile. Pooled EM is acceptable at this stage.
3. With $\mathbf{z}_m$ initialized, fit κ_{gc} and $\mathbf{v}_{gc}$ under the complete EC likelihood while holding $r_i = 0$.
4. Jointly fine-tune gene abundance, the program dictionaries P_g , program weights, and both primer observation models.
5. Fit candidate $r_i > 0$ models by adding small isoform-specific factors $\mathbf{h}_m, \mathbf{r}_{gc}$. Warm-start from smaller to larger r_i .

The fit should use Adam or another stable first-order optimizer initially, followed by deterministic full-data or large-batch polishing. Gene and isoform cell factors should use separate learning rates because isoform gradients are supported by fewer molecules. Gradient clipping and log-sum-exp evaluation are required for numerical stability.

Weak Gaussian priors or equivalent quadratic penalties on $\mathbf{h}_m$ and $\mathbf{r}_{gc}$ are appropriate even though the previous direct factorization did not use an L_2 penalty. Here they control an explicitly optional residual state that would otherwise fit sparse-cell noise. Their strength, or the decision to omit $\mathbf{h}_m$, must be selected from held-out molecules rather than labels.

Scalable Likelihood Evaluation

The likelihood constructs neither a dense cell-by-EC matrix nor a transcript-by-cell abundance block. For a cell minibatch, only ECs with nonzero counts contribute to $\sum_k n_{kmp} \log q_{kmp}$. Store D_{pkgc} sparsely by EC and gene-program pair. A COO event table containing cell, primer, EC, and

count joins each observed EC to those sparse projected program entries. Segmented reductions compute

$$q_{kmp} = \sum_{g,c} D_{pkgc} a_{gm} s_{gmc}.$$

The normalizing term uses the much smaller δ_{pgc} table:

$$\mathbf{1}' \mathbf{q}_{mp} = \sum_{g,c} \delta_{pgc} a_{gm} s_{gmc}.$$

For a minibatch of B cells, the main dense workspaces are $G \times B$ gene abundances and at most $(\sum_g C_g) \times B$ program weights. With small C_g , both are smaller than a $T \times B$ transcript block.

Because D_p depends on the learned P_g , it cannot remain permanently fixed while P_g is updated. The block-sparse matrix containing all P_g has only $\sum_g T_g C_g$ nonzeros. Sparse multiplication by A_p can refresh D_p after a global-program update, or gradients can flow through the sparse projection directly. A practical alternating schedule updates cell states and program weights for several steps with D_p cached, then updates P_g and refreshes D_p .

Explicit cell-factor storage remains small. For example, one million cells with a 32-dimensional float32 gene state require approximately 128 MB; an additional eight-dimensional isoform state requires approximately 32 MB. An amortized encoder from gene counts to $\mathbf{z}_m$ is a later option, but is not required for the initial million-cell implementation.

Identifiability

The following conventions remove avoidable invariances:

- Center the gene intercepts α_g , because a common shift of all gene log abundances is removed by the conditional multinomial likelihood.
- Constrain every column of P_g to the transcript simplex. Without this constraint, scale can move between P_g and a_{gm} .
- Center κ_{gc} , each coordinate of $\mathbf{v}_{gc}$, and each coordinate of $\mathbf{r}_{gc}$ over programs c within a gene. Adding a common value to every program logit leaves $\mathbf{s}_{gm}$ unchanged.
- Resolve program-label permutation for reporting by ordering programs by pooled weight and then matching warm-started programs to the preceding fit. Permutation does not change the likelihood.
- Center and standardize cell-factor columns after each epoch, absorbing location and scale changes into intercepts and loadings.
- Use factor balancing or an orthogonality convention for reporting. Latent rotations do not affect fitted abundance, but a stable convention improves warm starts and interpretation.

These constraints are parameter-identification conventions, not biological regularization and not simplex constraints on the genome-wide transcript matrix.

Label-Blind Model Selection

Molecule counts should be partitioned before any normalization, as in the existing count-fold procedure. Candidate models are compared using held-out multinomial log likelihood on the same response scale.

The recommended nested path for the independent isoform state is:

$$r_i = 0, 2, 4, 8, 16,$$

with small-to-large warm starts. The gene rank r_g can be selected first, then held fixed while selecting the small program cap $C_{\max}$, for example $C_{\max} = 1, 2, 3$. After selecting $C_{\max}$, a one-standard-error rule should prefer $r_i = 0$ or the smallest isoform rank statistically indistinguishable from the minimum held-out loss. A gene uses $C_g \leq \min(C_{\max}, T_g)$, and genes without evidence for variable isoform usage can remain at $C_g = 1$.

Held-out diagnostics should also decompose performance into:

- gene-unambiguous molecule likelihood;
- within-gene EC likelihood;
- cross-gene ambiguous EC likelihood;
- poly(dT) and random-hexamer likelihoods.

This distinguishes genuine improvement in isoform prediction from improvement that comes only through gene abundance. Cell-type labels, ARI, NMI, and silhouette remain external assessments and are never hyperparameter inputs.

Downstream Differential Isoform Analysis

The fitted $\pi_{gm} = P_g s_{gm}$ is the conditional isoform usage quantity needed downstream. Differential effects can be introduced in the program logits:

$$\lambda_{gmc} = \kappa_{gc} + \mathbf{v}'_{gc} \mathbf{z}_m + \mathbf{r}'_{gc} \mathbf{h}_m + \mathbf{x}'_m \boldsymbol{\gamma}_{gc},$$

where $\mathbf{x}_m$ contains diagnosis, cell type, donor covariates, and interactions. Coefficients $\boldsymbol{\gamma}_{gc}$ are centered over programs within each gene. Program-weight contrasts imply transcript-level changes through P_g , avoiding independent transcript effects that sparse cells cannot support. Inference should respect donor replication, either through donor-level random effects or by fitting and testing on donor-by-cell-type pseudobulks of the model-implied sufficient statistics.

This preserves the intended division of labor: gene factors provide a robust cell representation, shared isoform programs borrow information across cells, and downstream tests operate on transcript-resolved conditional usage rather than requiring independent cell-level EM estimates.

Recommended First Implementation

The first implementation should deliberately omit the free isoform cell state:

$$\log a_{gm} = \alpha_g + \mathbf{u}'_g \mathbf{z}_m, \quad \pi_{gm} = P_g s_{gm}, \quad s_{gmc} = \text{softmax}_c(\kappa_{gc} + \mathbf{v}'_{gc} \mathbf{z}_m),$$

with

$$q_{kmp} = \sum_{g,c} D_{pkgc} a_{gm} s_{gmc}.$$

It should use $C_{\max} = 2$, the paired-primer multinomial EC likelihood, pooled-EM initialization of one P_g column, explicit cell factors, and GPU minibatches. The fit evaluates the likelihood through D_p and δ_p , never through a materialized Θ . Only after this model is converged and evaluated should an independent $\mathbf{h}_m$ be added. This ordering tests the central hypothesis—that robust gene state can support transcript-resolution inference—without giving every sparse cell enough freedom to reproduce its EC noise.